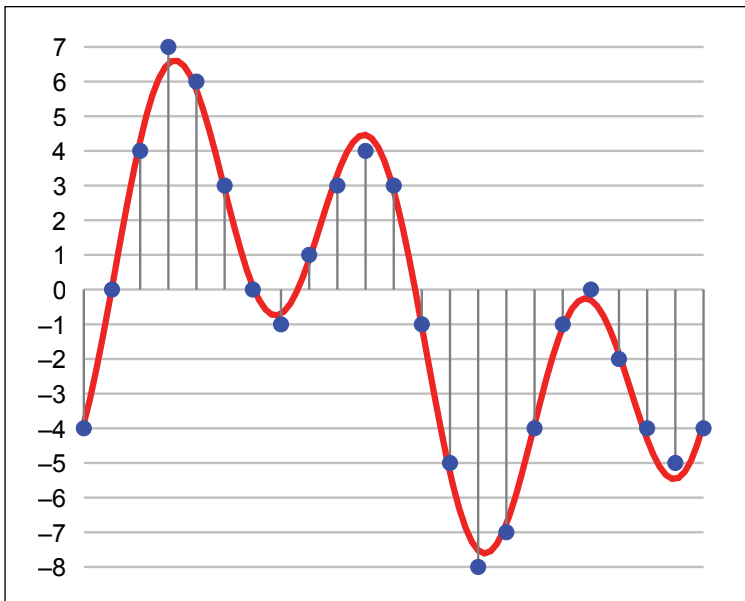


signal, it is 'sampled' at a particular periodic interval. Sampling simply means that the instantaneous value of the continuously varying signal is checked and assigned to one of the available discrete values. The number of values available to select from depends on the bit depth of the sampling ADC. For CD quality audio, the bit depth is 16 bits per channel (right and left). In the graph, this is the number of dashed black lines parallel to the time axis. Each dashed line corresponds to one of the discrete digital values that the signal can take. It is visually apparent that a higher density of dashed lines would mean more sensitivity and precision with regard to the values, as the digital value would be closer to the instantaneous analog value. Similar to the common video term, the bit depth can also be called the resolution of the audio.

As there is a time component, an ADC also has a clock in its IC which determines the speed of the sampling. The rate of sampling is called the sampling frequency, and is given by the number of samples per unit second. Thus, the sampling rate is a value with the unit in Hz. This value is 44KHz for CD quality audio, meaning 44000 samples are taken from the analog signal in one second, and each sample has a size of 16 bits (for one channel). In the Illustrative Graph, each red line corresponds to a single sample.



A 4-bit ADC sampling.. Better, but still not good enough